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MILITARY SPECIFICATION

PUMP, CENTRIFUGAL, GASOLINE ENGINE DRIVEN PORTABLE, FIRE FIGHTING SERVICE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers a portable fire pump which shall be self-contained, centrifugal, and gasoline engine driven.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-N-286 — Nickel-Copper-Aluminum Alloy, Wrought (K-Monel).

MILITARY

MIL-I-15024 — Identification Plates, Information Plates and Marking Information for Identification of Electrical, Electronic and Mechanical Equipment.

MIL-T-15071 — Technical Manuals for Mechanical and Electrical Equipment.

MIL-R-15137 — Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).

MIL-A-15153 — Aluminum Base Alloy Die Castings.

MIL-I-16165 — Interference Shielding, Engine Systems.

MIL-E-16400 — Electronic Equipment, Naval Ship and Shore: General Specification.

MIL-M-16576 — Metal Gun: Castings.

MIL-P-16789 — Preservation, Packaging, Packing and Marking of Pumps General, and Associated Repair Parts.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

FSC 4320

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2.2 **Other publications.** The following document forms a part of this specification. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATION

Handbook H28 — Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 **Qualification.** Pumps furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein (see 6.2).

3.2 **Dimensions.** The pump, engine, priming unit, and all appurtenances required to form a complete unit shall be mounted within a tubular aluminum protecting frame having approximate overall dimensions as follows:

24 inches long, 21 inches wide, and 20 inches high.

3.3 **Weight.** The complete unit less fuel tanks, hoses, and foot valve shall weigh approximately 140 pounds.

3.4 The pump shall be directly connected to the engine. The priming of the centrifugal pump shall be automatic by a vacuum pump driven from the engine and capable of priming with a suction lift of 20 feet.

3.5 The pump shall be rated at 250 gallons per minute (g. p. m.) of sea water (specific gravity 1.03) at a discharge pressure of 100 pounds per square inch (p.s.i.) and suction lift of 16 feet.

3.6 **Connections.** Pump suction and discharge connections shall be provided with hose thread conforming to Handbook H28. The suction connection shall be of 3-inch diameter and the discharge connection of 2½-inch diameter provided with a 3-way gate valve with one 2½-inch female and two 1½-inch male connections and one 2½-inch male connection. Each unit shall be furnished with a reducing adaptor having a 2½-inch female connection and a 1½-inch male connection.

3.6.1 Three lengths, 10 feet each, of noncollapsible, lightweight, oil-resisting, rubber covered, 3-inch inside diameter (I.D.) suction hose shall be furnished with each pump. Suitable hose couplings shall also be provided.

3.6.2 An exhaust hose 2-inch I.D. and 20 feet long, noncollapsible, lightweight and synthetic rubber covered shall also be provided.

3.6.3 Suction and discharge connections shall be arranged so there will be no unnecessary bends in the hose when attached to the pump and lying in a horizontal plane.

3.7 The unit shall be provided with a suitable foot valve with screening. Handles for easy handling and lashing shall be attached to the foot valve.

3.8 Pump casing shall be made of aluminum alloy in accordance with Specification MIL–A–15153, and shall be designed to withstand a hydrostatic pressure 150 percent of the pump shut-off pressure.

3.8.1 Pump casing shall be provided with a suitable opening for purpose of priming from external source at lifts higher than 20 feet. This opening shall be provided with means for quick opening and closing.

3.8.2 Suitable means shall be provided for draining pump casing and engine to prevent freezing during cold weather.

3.9 Impeller shall be gun metal in accordance with Specification MIL–M–16576. Each impeller shall be statically balanced.

3.10 Pump shaft shall be nickel-copper-aluminum alloy in accordance with Specification QQ–N–286.

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3.11 Pump casing and impeller shall be readily accessible for inspection and repairs.

3.12 **Engine.** The engine shall be designed for operation with any commercial grade of gasoline. A rewinding rope starter shall be provided as a permanent part of the unit. Suitable means shall be provided for emergency starting.

3.12.1 The unit shall be provided with a water pressure gage having a range of 0 to 200 p. s. i. and shall be shock-mounted and dampened against water pulsations. Suitable governing devices shall be installed to prevent overspeed of the engine in event of loss of prime.

3.12.2 A fuel tank of sufficient capacity to operate the engine at full power for a minimum of 2 hours shall be furnished. The tank shall be equipped with a suitable level indicator and a carrying handle. A hose to connect the tank to the engine shall be provided.

3.13 **Identification plates.** Each unit shall be provided with a brass instruction plate, securely attached and containing instructions for mixing oil and gasoline and the proper sequence for starting the engine. A brass identification plate shall be provided. Both identification plates shall be in accordance with Specification MIL–I–15024.

3.14 Each unit shall be radio shielded in accordance with Specification MIL–E–16400. The pump shall meet the radio interference requirements specified in Specification MIL–I–16165.

3.15 **Repair parts and tools.** Unless otherwise specified in the contract or order, repair part and tools shall be furnished as follows:

3.15.1 *Onboard.* Onboard repair parts shall be furnished and processed in accordance with Specification MIL–R–15137.

3.15.2 Each set of repair parts shall be based on 1 to 5 complete pumping units and shall consist of all wearing parts that require replacement under ordinary service conditions and special tools (see 6.3), as applicable, that are considered necessary by the bureau or agency concerned and the manufacturer.

3.16 **Master drawings.** Drawings shall be furnished as specified (see 6.1).

3.17 **Technical manuals.** Technical manuals shall be furnished as specified (see 6.1). Unless otherwise specified (see 6.1), manuals shall be type C in accordance with Specification MIL–T–15071.

3.18 **Workmanship.** The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Classification of tests.** Tests shall be classified as follows:

(a) Qualification tests.

(b) Contract acceptance tests.

(c) Hydrostatic tests.

4.2 **Qualification tests at a Government laboratory.** Qualification tests shall be conducted at a Government laboratory designated by the Bureau of Ships. These tests shall be conducted to determine that performance and design of the pumps are in compliance with this specification. Minor deviations in design and materials may be allowed at the discretion of the Bureau of Ships.

4.2.1 The following tests shall be made to determine qualification:

(a) Performance tests to demonstrate the ability of the pump to operate satisfactorily at suction lifts of 3 feet, 16 feet, and 22 feet without an eductor; and at 50 feet using a suitable eductor. Sufficient data shall be taken during these tests to prepare pump characteristic curves.

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(b) Cold starting trials at the following conditions:

1. Ten starts at 0°F.
2. Ten starts at minus 10°F.
3. Ten starts at minus 20°F.

(c) The endurance tests shall consist of 200 hours of running time. This operation shall be a series of intermittent runs of approximately 3 hours each at rated capacity, suction and discharge conditions, and speed. Failure of any part except spark plugs, packing or seals during the endurance test shall be considered as failure of the pump to qualify. The pump shall complete the endurance test without excessive noise, vibration or wear of parts.

(d) Radio interference tests to determine compliance with Specification MIL–I–16165.

4.3 Contract acceptance tests. Contract acceptance tests shall be conducted at the place of manufacture, shall be witnessed by the Government inspector, and shall consist of the following:

(a) Routine tests — required on each pump.

(b) Selective tests — required on selected pumps from a production run.

4.3.1 *Routine tests.* Each unit shall be operated for one hour minimum. For at least 5 minutes of this period, the unit shall be operated at rated capacity and at rated suction lift or simulated suction lift.

4.3.2 *Selective tests.*

4.3.2.1 *Lot.* A lot shall consist of a group of pumps of the same design produced under uniform conditions which have successfully passed the routine tests.

4.3.2.2 *Sampling.* Pumps shall be selected by or under the supervision of Government inspector for selective tests in accordance with table I.

TABLE I. *Sampling for selective tests*

Number of pumps in lot	Minimum number of pumps to be tested
15 and under	3
16 to 40	5
41 to 110	7
111 to 300	10
301 to 500	15
501 and over	1 out of each 20 produced

4.3.2.3 *Tests.* Sample pumps selected in accordance with 4.3.2.2 and table I shall be operated for a period of 4 hours at rated capacity and at rated suction lift or simulated suction lift in addition to and after completion of the routine tests. If any sample pump is found not to conform to this specification, the lot which it represents shall be rejected. A rejected lot may be resubmitted for selective tests provided the manufacturer, after having been informed of the reasons for rejection, has inspected or tested each pump in the lot for deficiencies noted and has removed all nonconforming units.

4.4 Hydrostatic tests. Each pump casing shall be subjected to a hydrostatic test of two times the working pressure of 100 pounds per square inch.

5. PREPARATION FOR DELIVERY

5.1 The pump and repair parts shall be cleaned, preserved, packaged, packed and marked in accordance with Specification MIL–P–16789.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

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(a) Title, number, and date of this specification.

(b) Applicable level of packaging and packing required (see 5.1).

(c) Repair parts and tools required if other than as specified (see 3.15).

(d) Drawings required (see 3.16).

(e) Technical manuals required (see 3.17).

6.2 **Qualification.** With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL–16359, whether or not such products have actually been so listed by that date.

6.2.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D. C.

6.3 Special tools are defined as those tools not listed in the Catalog of Naval Material, General Stores Section. (Copies of this catalog may be consulted in the office of the Government inspector.)

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Navy—Bureau of Ships
Air Force

Other interest:
Army—Sig

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